Question	Answer	Marks
6(a)(i)	non-zero horizontal straight line from X to Y	B1
6(a)(ii)	constant flux density (inside coil)	B1
	either (magnetic) flux linkage proportional to flux density **or** $\Phi = BAN$ and B, A and N are all constant	B1
6(a)(iii)	$\Phi = BAN$	C1
	$= 0.080 \times 0.71 \times 10^{-4} \times 64$ $= 3.6 \times 10^{-4}$ Wb	A1
6(a)(iv)	*sketch showing*: E is zero from time 0 to time t and non-zero after time t	B1
	E has constant non-zero magnitude between time t and time $4t$	B1
	E has non-zero value of one sign between time t and time $2t$, and non-zero value of the opposite sign between time $2t$ and time $4t$	B1
6(b)	current in spring creates a magnetic field around the spring	B1
	either (magnetic) fields around adjacent turns interact to cause a force to be exerted (between the turns) **or** current in one turn interacts with (magnetic) field due to adjacent turns to cause force to be exerted (between the turns)	B1
	(magnetic force) is attractive so distance (between turns) decreases	B1

Question	Answer	Marks